Differential Equation: Questions and Full Solutions

Question 01

English

Let $\mathcal{F}$ be the family of all tangent lines to the curve $y = e^x$, $x \in \mathbb{R}$. With $p = \dfrac{dy}{dx}$, the differential equation of $\mathcal{F}$, free of every arbitrary constant, is

(A) $y = p(x-1) + p\ln p$
(B) $y = px - p\ln p$
(C) $y = p(x+1) - p\ln p$
(D) $y = p(x+1) + p\ln p$

Only one option is correct.

हिन्दी

माना कि $\mathcal{F}$ वक्र $y = e^x$, $x \in \mathbb{R}$ की सभी स्पर्श रेखाओं का कुल है। $p = \dfrac{dy}{dx}$ लेने पर, $\mathcal{F}$ का प्रत्येक स्वेच्छ अचर से मुक्त अवकल समीकरण है

(A) $y = p(x-1) + p\ln p$
(B) $y = px - p\ln p$
(C) $y = p(x+1) - p\ln p$
(D) $y = p(x+1) + p\ln p$

केवल एक विकल्प सही है।

Solution

The point of contact on $y = e^x$ is (t, e^t), $t \in \mathbb{R}$, and the slope there is e^t.

Tangent line: $y - e^t = e^t(x - t)$, that is, $y = e^t(x - t + 1)$. Each member of $\mathcal{F}$ corresponds to one value of t, so the family has one arbitrary constant and its differential equation is of first order.

Differentiate the equation of the line with respect to x (with t fixed): $\dfrac{dy}{dx} = e^t$, so $p = e^t$ and $t = \ln p$ (valid since $p > 0$).

Substitute in the line: $y = p\,(x - \ln p + 1) = p(x+1) - p\ln p$.

Check with the tangent at $t = 0$: $y = x + 1$, $p = 1$, and $p(x+1) - p\ln p = x + 1$, as required.

Option (B) comes from writing the tangent as $y = e^t(x - t)$; option (D) from a sign slip in $-pt = -p\ln p$; option (A) from using $x + t - 1$.

Correct Option: (C)

Question 02

English

Consider the differential equation $x\dfrac{dy}{dx} = 2y$, $x \in \mathbb{R}$, whose general solution is usually written as $y = cx^2$, $c \in \mathbb{R}$. Which of the following functions $y : \mathbb{R} \to \mathbb{R}$ is differentiable on $\mathbb{R}$, satisfies the equation at every $x \in \mathbb{R}$, and is not of the form $y = cx^2$ for any single constant c?

(A) $y = \begin{cases} x^2, & x \geq 0 \\ -2x^2, & x < 0 \end{cases}$

(B) $y = 3x^2$

(C) $y = \begin{cases} x^2, & x \geq 0 \\ x^2 + x, & x < 0 \end{cases}$

(D) $y = |x|^3$

Only one option is correct.

हिन्दी

अवकल समीकरण $x\dfrac{dy}{dx} = 2y$, $x \in \mathbb{R}$ पर विचार कीजिए, जिसका व्यापक हल प्रायः $y = cx^2$, $c \in \mathbb{R}$ के रूप में लिखा जाता है। निम्न में से कौन-सा फलन $y : \mathbb{R} \to \mathbb{R}$, $\mathbb{R}$ पर अवकलनीय है, प्रत्येक $x \in \mathbb{R}$ पर समीकरण को संतुष्ट करता है, तथा किसी भी एक अचर c के लिए $y = cx^2$ के रूप का नहीं है?

(A) $y = \begin{cases} x^2, & x \geq 0 \\ -2x^2, & x < 0 \end{cases}$

(B) $y = 3x^2$

(C) $y = \begin{cases} x^2, & x \geq 0 \\ x^2 + x, & x < 0 \end{cases}$

(D) $y = |x|^3$

केवल एक विकल्प सही है।

Solution

Separating variables, $\dfrac{dy}{y} = \dfrac{2\,dx}{x}$ on any interval not containing 0 (where $y \neq 0$), so on $(0, \infty)$ every solution is $y = c_1x^2$ and on $(-\infty, 0)$ every solution is $y = c_2x^2$ (the zero function is included as $c = 0$).

At $x = 0$ the equation reads $0 \cdot y'(0) = 2y(0)$, so it only requires $y(0) = 0$, which both pieces give. A function made of c_1x^2 for $x \geq 0$ and c_2x^2 for $x < 0$ has right derivative 0 and left derivative 0 at the origin, so it is differentiable on $\mathbb{R}$. Hence the full set of solutions on $\mathbb{R}$ is a two-parameter set, larger than the one-parameter family $y = cx^2$.

(A) is this construction with $c_1 = 1$, $c_2 = -2$. It satisfies the equation for every x and is differentiable at 0. Since $c_1 \neq c_2$, no single c gives it. So (A) satisfies all three conditions.

(B) $y = 3x^2$ is a solution, but it is cx^2 with $c = 3$, so it fails the last condition.

(C) For $x < 0$: $xy' = x(2x + 1) = 2x^2 + x$ while $2y = 2x^2 + 2x$; these differ, so the equation fails. Moreover the left derivative at 0 is 1 and the right derivative is 0, so it is not differentiable at 0.

(D) For $x > 0$: $y = x^3$, $xy' = 3x^3 \neq 2x^3 = 2y$, so it is not a solution.

Correct Option: (A)

Question 03

English

Let $f : (0, \infty) \to \mathbb{R}$ be a differentiable function such that $xf'(x) - f(x) = x^2 (1 + \ln x)$ for all $x > 0$ and $\lim\limits_{x \to 0^+} \dfrac{f(x)}{x} = 1$. Then $f(e)$ is equal to

(A) e^2
(B) $e^2 + e$
(C) $2e^2 + e$
(D) $e^2 - e$

Only one option is correct.

हिन्दी

माना कि $f : (0, \infty) \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए सभी $x > 0$ हेतु $xf'(x) - f(x) = x^2 (1 + \ln x)$ तथा $\lim\limits_{x \to 0^+} \dfrac{f(x)}{x} = 1$ है। तब $f(e)$ बराबर है

(A) e^2
(B) $e^2 + e$
(C) $2e^2 + e$
(D) $e^2 - e$

केवल एक विकल्प सही है।

Solution

Write the equation as $f'(x) - \dfrac{1}{x} f(x) = x(1 + \ln x)$, a linear equation with $P = -\dfrac{1}{x}$. The integrating factor is $e^{-\int dx/x} = \dfrac{1}{x}$ on $(0, \infty)$.

Multiplying by the integrating factor is the same as dividing the original equation by x^2, and the left side is then an exact derivative: $\dfrac{xf'(x) - f(x)}{x^2} = \dfrac{d}{dx}\left(\dfrac{f(x)}{x}\right)$.

The right side becomes $1 + \ln x$. Since $\dfrac{d}{dx}(x \ln x) = \ln x + 1$, it is also an exact derivative.

Hence $\dfrac{d}{dx}\left(\dfrac{f(x)}{x} - x \ln x\right) = 0$ on $(0, \infty)$, so $\dfrac{f(x)}{x} = x \ln x + C$.

Now $\lim\limits_{x \to 0^+} x \ln x = 0$, so $\lim\limits_{x \to 0^+} \dfrac{f(x)}{x} = C$. The given limit is 1, so $C = 1$.

Therefore $f(x) = x^2 \ln x + x$ and $f(e) = e^2 \cdot 1 + e = e^2 + e$.

Option (C) results from integrating $1 + \ln x$ as $x + x \ln x$ (forgetting that $\int \ln x \, dx = x \ln x - x$); option (A) from taking $C = 0$.

Correct Option: (B)

Question 04

English

Let C be the curve in the region $y > 0$ passing through the point $(0, 1)$ such that every point (x, y) of C satisfies $y\,dx = \left(x + 2y^3e^y\right) dy$. If C meets the line $y = 2$ at the point $(\alpha, 2)$, then α is equal to

(A) $2e^2$
(B) $8e^2$
(C) e^2
(D) $4e^2$

Only one option is correct.

हिन्दी

माना कि C क्षेत्र $y > 0$ में बिन्दु $(0, 1)$ से होकर जाने वाला वह वक्र है जिसका प्रत्येक बिन्दु (x, y) समीकरण $y\,dx = \left(x + 2y^3e^y\right) dy$ को संतुष्ट करता है। यदि C रेखा $y = 2$ से बिन्दु $(\alpha, 2)$ पर मिलता है, तो α बराबर है

(A) $2e^2$
(B) $8e^2$
(C) e^2
(D) $4e^2$

केवल एक विकल्प सही है।

Solution

Solving for $\dfrac{dy}{dx}$ gives $\dfrac{dy}{dx} = \dfrac{y}{x + 2y^3e^y}$, which is not linear in y. Treat x as a function of y instead: dividing by $y\,dy$ $(y > 0)$, $\dfrac{dx}{dy} = \dfrac{x}{y} + 2y^2e^y$, that is, $\dfrac{dx}{dy} - \dfrac{1}{y}x = 2y^2e^y$.

This is linear in x with $P(y) = -\dfrac{1}{y}$. Integrating factor: $e^{-\int dy/y} = \dfrac{1}{y}$.

Multiply: $\dfrac{d}{dy}\left(\dfrac{x}{y}\right) = 2ye^y$. Integrating by parts, $\int 2ye^y dy = 2ye^y - 2e^y + C$, so $\dfrac{x}{y} = 2(y-1)e^y + C$.

The curve passes through $(0, 1)$: $\dfrac{0}{1} = 2(0)e + C$, so $C = 0$ and $x = 2y(y - 1)e^y$.

For $y = 2$: $\alpha = 2 \cdot 2 \cdot (2 - 1)e^2 = 4e^2$.

Option (A) comes from forgetting to multiply $\dfrac{x}{y}$ back by y.

Correct Option: (D)

Question 05

English

Let C be a curve in the first quadrant passing through the point $(1, 2)$ such that, for every point $P(x, y)$ on C, the normal to C at P meets the x-axis at the point $(3x, 0)$. If the point $\left(\sqrt{7}, \beta\right)$ lies on C, then β is equal to

(A) 4
(B) 3
(C) $\sqrt{15}$
(D) $2\sqrt{3}$

Only one option is correct.

हिन्दी

माना कि C प्रथम चतुर्थांश में बिन्दु $(1, 2)$ से होकर जाने वाला एक वक्र है, जिसके प्रत्येक बिन्दु $P(x, y)$ के लिए P पर C का अभिलम्ब x-अक्ष से बिन्दु $(3x, 0)$ पर मिलता है। यदि बिन्दु $\left(\sqrt{7}, \beta\right)$, C पर स्थित है, तो β बराबर है

(A) 4
(B) 3
(C) $\sqrt{15}$
(D) $2\sqrt{3}$

केवल एक विकल्प सही है।

Solution

At $P(x, y)$ the normal is $Y - y = -\dfrac{1}{y'}(X - x)$. Putting $Y = 0$: $X = x + yy'$. So the normal meets the x-axis at $(x + yy', 0)$; the length yy' is the subnormal.

The condition gives $x + yy' = 3x$, that is, $y\dfrac{dy}{dx} = 2x$.

Separating: $2y\,dy = 4x\,dx \Rightarrow y^2 = 2x^2 + C$.

Since $(1, 2)$ lies on C: $4 = 2 + C$, so $C = 2$ and C is the part of $y^2 = 2x^2 + 2$ in the first quadrant.

At $x = \sqrt{7}$: $y^2 = 14 + 2 = 16$, so $\beta = 4$ (positive, since C is in the first quadrant).

Option (D) comes from using the tangent instead of the normal or taking $yy' = x$; option (B) from $y^2 = x^2 + 2$.

Correct Option: (A)

Question 06

English

In a town of N people a rumour spreads in such a way that the rate of increase of the number $R(t)$ of people who have heard it is proportional to the product of $R(t)$ and the number of people who have not heard it, t being the time in hours. At $t = 0$, $R = \dfrac{N}{5}$, and at $t = 2$, $R = \dfrac{N}{2}$. Then $R(3)$ is equal to

(A) $\dfrac{3N}{4}$

(B) $\dfrac{2N}{3}$

(C) $\dfrac{4N}{5}$

(D) $\dfrac{5N}{8}$

Only one option is correct.

हिन्दी

N लोगों के एक नगर में एक अफवाह इस प्रकार फैलती है कि उसे सुन चुके लोगों की संख्या $R(t)$ के बढ़ने की दर, $R(t)$ तथा उसे न सुनने वाले लोगों की संख्या के गुणनफल के समानुपाती है, जहाँ t घंटों में समय है। $t = 0$ पर $R = \dfrac{N}{5}$ तथा $t = 2$ पर $R = \dfrac{N}{2}$ है। तब $R(3)$ बराबर है

(A) $\dfrac{3N}{4}$

(B) $\dfrac{2N}{3}$

(C) $\dfrac{4N}{5}$

(D) $\dfrac{5N}{8}$

केवल एक विकल्प सही है।

Solution

The model is $\dfrac{dR}{dt} = kR(N - R)$ with $0 < R < N$, $k > 0$ a constant.

Separate the variables: $\dfrac{dR}{R(N-R)} = k\,dt$. Partial fractions: $\dfrac{1}{R(N-R)} = \dfrac{1}{N}\left(\dfrac{1}{R} + \dfrac{1}{N-R}\right)$.

Integrate: $\dfrac{1}{N}\left(\ln R - \ln(N-R)\right) = kt + c_1$, that is, $\ln \dfrac{R}{N-R} = kNt + c$.

At $t = 0$, $R = \dfrac{N}{5}$, so $\dfrac{R}{N-R} = \dfrac{1}{4}$ and $c = -\ln 4$.

At $t = 2$, $R = \dfrac{N}{2}$, so $\dfrac{R}{N-R} = 1$: $0 = 2kN - \ln 4$, hence $kN = \ln 2$.

So $\dfrac{R}{N-R} = \dfrac{1}{4} \cdot 2^t$. The ratio doubles every hour: $\dfrac{1}{4}, \dfrac{1}{2}, 1, 2$ at $t = 0, 1, 2, 3$.

At $t = 3$: $\dfrac{R}{N-R} = 2 \Rightarrow R = 2N - 2R \Rightarrow R = \dfrac{2N}{3}$.

Option (D) is the linear interpolation; option (C) assumes the curve is symmetric about $t = 2$ after one hour.

Correct Option: (B)

Question 07

English

Let $y : (0, \infty) \to \mathbb{R}$ be a twice differentiable function such that $x\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} = 4x$ for all $x > 0$, $y(1) = 2$ and $y(e) = e^2 + 4$. Then $y\left(e^2\right)$ is equal to

(A) $e^4 + 5$
(B) $e^4 + 6$
(C) $e^4 + 8$
(D) $e^4 + 7$

Only one option is correct.

हिन्दी

माना कि $y : (0, \infty) \to \mathbb{R}$ एक दो बार अवकलनीय फलन है, जिसके लिए सभी $x > 0$ हेतु $x\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} = 4x$, $y(1) = 2$ तथा $y(e) = e^2 + 4$ है। तब $y\left(e^2\right)$ बराबर है

(A) $e^4 + 5$
(B) $e^4 + 6$
(C) $e^4 + 8$
(D) $e^4 + 7$

केवल एक विकल्प सही है।

Solution

The left side is an exact derivative: $\dfrac{d}{dx}\left(x\dfrac{dy}{dx}\right) = x\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx}$. So $\dfrac{d}{dx}\left(xy'\right) = 4x$.

Integrate: $xy' = 2x^2 + a$, so $y' = 2x + \dfrac{a}{x}$ for $x > 0$.

Integrate again: $y = x^2 + a\ln x + b$. There are two arbitrary constants, so two conditions are needed; here they are given at two different points (a boundary value problem).

$y(1) = 1 + 0 + b = 2$ gives $b = 1$.

$y(e) = e^2 + a + 1 = e^2 + 4$ gives $a = 3$.

So $y = x^2 + 3\ln x + 1$ and $y\left(e^2\right) = e^4 + 3 \cdot 2 + 1 = e^4 + 7$.

Option (B) omits the constant b; option (A) uses $\ln e^2 = 1$.

Correct Option: (D)

Question 08

English

Let m and n denote the order and the degree, respectively, of the differential equation

$$\left(\frac{d^2y}{dx^2}\right)^{5/3} = \left(1 + \left(\frac{dy}{dx}\right)^2\right)^{1/2}.$$

Then the value of $m - n$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि अवकल समीकरण

$$\left(\frac{d^2y}{dx^2}\right)^{5/3} = \left(1 + \left(\frac{dy}{dx}\right)^2\right)^{1/2}$$

की कोटि तथा घात क्रमशः m तथा n हैं। तब $m - n$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The order is the order of the highest derivative present, $\frac{d^2y}{dx^2}$, so $m = 2$.

The degree is defined only after the equation is made a polynomial in the derivatives. The exponents are $\frac{5}{3}$ and $\frac{1}{2}$; the least common multiple of the denominators is 6.

Raising both sides to the power 6: $\left(\frac{d^2y}{dx^2}\right)^{10} = \left(1 + \left(\frac{dy}{dx}\right)^2\right)^3$. Now both sides are polynomials in y' and y''.

The highest-order derivative y'' appears with power 10, so $n = 10$. (Raising only to the power 3 would leave a square root and give the wrong value 5.)

Hence $m - n = 2 - 10 = -8$.

Final Answer: -8

Question 09

English

Let m and n denote the order and the degree, respectively, of the differential equation of the family of curves

$$y = \sum_{k=1}^{10} \left(a_k \sin^2 kx + b_k \cos^2 kx\right),$$

where $a_1, a_2, \ldots, a_{10}, b_1, b_2, \ldots, b_{10}$ are arbitrary constants. Then the value of $m + n$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि वक्र-कुल

$$y = \sum_{k=1}^{10} \left(a_k \sin^2 kx + b_k \cos^2 kx\right),$$

जहाँ $a_1, a_2, \ldots, a_{10}, b_1, b_2, \ldots, b_{10}$ स्वेच्छ अचर हैं, के अवकल समीकरण की कोटि तथा घात क्रमशः m तथा n हैं। तब $m + n$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Use $\sin^2 kx = \frac{1 - \cos 2kx}{2}$ and $\cos^2 kx = \frac{1 + \cos 2kx}{2}$.

Then $y = \frac{1}{2}\sum_{k=1}^{10}(a_k + b_k) + \sum_{k=1}^{10} \frac{b_k - a_k}{2} \cos 2kx = A_0 + \sum_{k=1}^{10} A_k \cos 2kx$.

The 20 constants enter only through the 11 combinations $A_0, A_1, \ldots, A_{10}$, and these can take any real values. The functions $1, \cos 2x, \cos 4x, \ldots, \cos 20x$ are linearly independent (for example, by

orthogonality on $[0, \pi]$), so none of the 11 constants can be absorbed into the others: the family has exactly 11 essential constants.

The differential equation of a family with 11 essential constants has order 11, so $m = 11$.

The family is a linear combination of fixed functions, so eliminating the constants (by the Wronskian of $y, 1, \cos 2x, \ldots, \cos 20x$) gives an equation linear in $y, y', \ldots, y^{(11)}$. Its degree is 1, so $n = 1$.

Hence $m + n = 12$. Counting all 20 constants would give the wrong order 20.

Final Answer: 12

Question 10

English

For $a > 0$, let S_a denote the circle $(x - a)^2 + y^2 = \dfrac{a^2}{2}$, and let $\mathcal{F} = \{S_a : a > 0\}$. Exactly two members of $\mathcal{F}$ pass through the point $P(5, 2)$. If m_1 and m_2 are the slopes of the tangents to these two circles at P, then the value of $m_1 m_2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$a > 0$ के लिए माना कि S_a वृत्त $(x - a)^2 + y^2 = \dfrac{a^2}{2}$ को दर्शाता है, तथा $\mathcal{F} = \{S_a : a > 0\}$ है। $\mathcal{F}$ के ठीक दो सदस्य बिन्दु $P(5, 2)$ से होकर जाते हैं। यदि P पर इन दो वृत्तों की स्पर्श रेखाओं की प्रवणताएँ m_1 तथा m_2 हैं, तो $m_1 m_2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Differentiate the equation of S_a with respect to x: $2(x-a)+2y\dfrac{dy}{dx} = 0$, so with $p = \dfrac{dy}{dx}$, $x - a = -yp$ and $a = x + yp$.

Substitute in $(x - a)^2 + y^2 = \dfrac{a^2}{2}$: $y^2p^2 + y^2 = \dfrac{(x + yp)^2}{2}$.

Multiply by 2 and expand: $2y^2p^2 + 2y^2 = x^2 + 2xyp + y^2p^2$, so $y^2p^2 - 2xyp + 2y^2 - x^2 = 0$. This is the differential equation of the family; it is quadratic in p, matching the fact that two members can pass through one point.

At $P(5, 2)$: $4p^2 - 20p + 8 - 25 = 0$, that is, $4p^2 - 20p - 17 = 0$. Its roots are the slopes m_1, m_2 of the members through P.

By Vieta: $m_1 m_2 = -\dfrac{17}{4} = -4.25$.

Check that both members are admissible: $(5 - a)^2 + 4 = \dfrac{a^2}{2}$ gives $a^2 - 20a + 58 = 0$, $a = 10 \pm \sqrt{42}$, both positive. The slopes $\dfrac{a - 5}{2}$ give the same product $\dfrac{(a_1 - 5)(a_2 - 5)}{4} = \dfrac{58 - 100 + 25}{4} = -\dfrac{17}{4}$.

Final Answer: -4.25

Question 11

English

Let $y : [0, \infty) \to [0, \infty)$ be a differentiable function such that $\frac{dy}{dx} = 2\sqrt{y}$ for all $x \geq 0$, $y(0) = 0$ and $y(2) = 1$. Then the value of $y\left(\frac{3}{2}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $y : [0, \infty) \to [0, \infty)$ एक अवकलनीय फलन है, जिसके लिए सभी $x \geq 0$ हेतु $\frac{dy}{dx} = 2\sqrt{y}$, $y(0) = 0$ तथा $y(2) = 1$ है। तब $y\left(\frac{3}{2}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

On any interval where $y > 0$ the equation separates: $\frac{dy}{2\sqrt{y}} = dx$, so $\sqrt{y} = x - c$ and $y = (x - c)^2$, valid only where $x - c > 0$ (since $\sqrt{y} \geq 0$). This is the general solution. The zero function $y \equiv 0$ also satisfies the equation but is not obtained from it for any c (a singular solution).

If we used only the general solution, $y(0) = 0$ would force $c = 0$ and $y = x^2$, contradicting $y(2) = 1$. So the required solution must use both kinds of solution.

Since $y' = 2\sqrt{y} \geq 0$, y is non-decreasing. Let $c = \sup\{x : y(x) = 0\}$. Then $y = 0$ on $[0, c]$, and on (c, ∞), $y > 0$, so $y = (x - c')^2$ there with $\sqrt{y} = x - c'$; continuity at c gives $c' = c$. Hence $y = 0$ for $x \leq c$ and $y = (x - c)^2$ for $x \geq c$. At $x = c$ both one-sided derivatives are $0 = 2\sqrt{y(c)}$, so this spliced function is differentiable and is a solution for every $c \geq 0$.

$y(2) = 1$: if $c \geq 2$ then $y(2) = 0$, so $c < 2$ and $(2 - c)^2 = 1$, giving $c = 1$ (the root $c = 3$ is not less than 2).

So $y = 0$ on $[0, 1]$, $y = (x - 1)^2$ for $x \geq 1$, and $y\left(\frac{3}{2}\right) = \frac{1}{4} = 0.25$.

Final Answer: 0.25

Question 12

English

Let $f : (0, \infty) \to (0, \infty)$ be a differentiable function with $f(1) = 1$ such that $x^2 f'(x) = 3xf(x) - (f(x))^2$ for all $x > 0$. Let $g : (0, \infty) \to \mathbb{R}$ be a differentiable function with $g(1) = 0$ such that

$$g'(x) + \frac{2x}{1 + x^2} g(x) = f(x) \quad \text{for all } x > 0.$$

Then the value of $g(15)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : (0, \infty) \to (0, \infty)$ एक अवकलनीय फलन है, $f(1) = 1$, जिसके लिए सभी $x > 0$ हेतु $x^2 f'(x) = 3xf(x) - (f(x))^2$ है। माना कि $g : (0, \infty) \to \mathbb{R}$ एक अवकलनीय फलन है, $g(1) = 0$, जिसके लिए सभी $x > 0$ हेतु

$$g'(x) + \frac{2x}{1 + x^2} g(x) = f(x)$$

है। तब $g(15)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Divide the first equation by x^2: $f'(x) = 3\dfrac{f}{x} - \left(\dfrac{f}{x}\right)^2$. The right side depends only on $\dfrac{f}{x}$, so the equation is homogeneous.

Put $f = vx$, $f' = v + xv'$: $v + xv' = 3v - v^2$, so $x\dfrac{dv}{dx} = v(2 - v)$. With $f(1) = 1$, $v(1) = 1$, and v stays in $(0, 2)$ (the constant solutions $v = 0$ and $v = 2$ cannot be crossed).

Separate the variables: $\dfrac{dv}{v(2-v)} = \dfrac{dx}{x}$. Partial fractions: $\dfrac{1}{v(2-v)} = \dfrac{1}{2}\left(\dfrac{1}{v} + \dfrac{1}{2-v}\right)$, so $\dfrac{1}{2}\ln\dfrac{v}{2-v} = \ln x + c$.

At $x = 1$, $v = 1$: $c = 0$. Hence $\dfrac{v}{2-v} = x^2$, $v = \dfrac{2x^2}{1+x^2}$ and $f(x) = \dfrac{2x^3}{1+x^2}$, which is positive on $(0, \infty)$.

The second equation is linear in g with $P(x) = \dfrac{2x}{1+x^2}$. Integrating factor: $e^{\ln(1+x^2)} = 1 + x^2$.

Then $\dfrac{d}{dx}\left((1+x^2)g\right) = (1+x^2)f(x) = 2x^3$, so $(1+x^2)g(x) = \dfrac{x^4}{2} + K$.

$g(1) = 0$ gives $0 = \dfrac{1}{2} + K$, $K = -\dfrac{1}{2}$. So $g(x) = \dfrac{x^4 - 1}{2(1+x^2)} = \dfrac{(x^2-1)(x^2+1)}{2(x^2+1)} = \dfrac{x^2-1}{2}$.

$g(15) = \dfrac{225 - 1}{2} = 112$.

Final Answer: 112

Question 13

English

Let $y = y(x)$, $x > -1$, be the solution of the differential equation $\dfrac{dy}{dx} = \dfrac{2x - 2y - 1}{2x - 2y}$ with $y(0) = -1$. Then the value of $y\left(\dfrac{21}{4}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $y = y(x)$, $x > -1$, अवकल समीकरण $\dfrac{dy}{dx} = \dfrac{2x - 2y - 1}{2x - 2y}$ का वह हल है जिसके लिए $y(0) = -1$ है। तब $y\left(\dfrac{21}{4}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The lines $2x - 2y - 1 = 0$ and $2x - 2y = 0$ are parallel, so no shift of origin makes the equation homogeneous. Instead, the right side is a function of $x - y$ only: $\dfrac{dy}{dx} = \dfrac{2u - 1}{2u}$ with $u = x - y$.

Then $\dfrac{du}{dx} = 1 - \dfrac{dy}{dx} = 1 - \dfrac{2u-1}{2u} = \dfrac{1}{2u}$.

Separate: $2u\,du = dx$, so $u^2 = x + c$.

At $x = 0$: $u = 0 - (-1) = 1$, so $c = 1$ and $u^2 = x + 1$. Since $u(0) = 1 > 0$ and u cannot pass through 0 (the equation is undefined there), $u = \sqrt{x+1}$ on $x > -1$.

So $y = x - \sqrt{x+1}$. At $x = \frac{21}{4}$: $\sqrt{\frac{25}{4}} = \frac{5}{2}$, so $y = \frac{21}{4} - \frac{5}{2} = \frac{11}{4} = 2.75$.

Final Answer: 2.75

Question 14

English

Let $y = y(x)$, $0 \le x < 1$, be the solution of the differential equation $e^y \frac{dy}{dx} = (x + e^y)^2 - 1$ with $y(0) = 0$. Then the value of $e^{y(\frac{3}{4})}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $y = y(x)$, $0 \le x < 1$, अवकल समीकरण $e^y \frac{dy}{dx} = (x + e^y)^2 - 1$ का वह हल है जिसके लिए $y(0) = 0$ है। तब $e^{y(\frac{3}{4})}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Since $\frac{d}{dx}e^y = e^y \frac{dy}{dx}$, put $t = e^y$ $(t > 0)$. The equation becomes $\frac{dt}{dx} = (x + t)^2 - 1$.

The right side is a function of $x + t$ only, so put $u = x + t$: $\frac{du}{dx} = 1 + \frac{dt}{dx} = 1 + (x + t)^2 - 1 = u^2$.

Separate: $\frac{du}{u^2} = dx$, so $-\frac{1}{u} = x + c$.

At $x = 0$: $u = 0 + e^0 = 1$, so $c = -1$ and $u = \frac{1}{1-x}$ for $0 \le x < 1$ (the solution exists on this interval).

Then $e^y = t = u - x = \frac{1}{1-x} - x$, which is positive on $[0, 1)$ because $\frac{1}{1-x} \ge 1 > x$.

At $x = \frac{3}{4}$: $e^{y(3/4)} = 4 - \frac{3}{4} = \frac{13}{4} = 3.25$.

Final Answer: 3.25

Question 15

English

For a point $P(x, y) \neq (0, 0)$, let $\theta \in (-\pi, \pi]$ be the angle with $x = \sqrt{x^2 + y^2}\cos\theta$ and $y = \sqrt{x^2 + y^2}\sin\theta$. Let C be the curve $x^2 + y^2 = g(\theta)$, $-\pi < \theta < 1$, where g is a differentiable function with $g(0) = 1$, such that every point of C satisfies

$$2\left(x^2 + y^2\right)(x\,dx + y\,dy) = \left[\left(x^2 + y^2 + 2\theta\right)^2 - 2\right](x\,dy - y\,dx).$$

The curve C meets the circle $x^2 + y^2 = \frac{5}{3}$ at exactly two points. If $\theta_1 < \theta_2$ are the values of θ at these points, then the value of θ_1 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

बिन्दु $P(x, y) \neq (0, 0)$ के लिए माना कि $\theta \in (-\pi, \pi]$ वह कोण है जिसके लिए $x = \sqrt{x^2 + y^2}\cos\theta$ तथा $y =$

$\sqrt{x^2+y^2}\sin\theta$ है। माना कि C वक्र $x^2+y^2=g(\theta)$, $-\pi<\theta<1$ है, जहाँ g एक अवकलनीय फलन है, $g(0)=1$, तथा C का प्रत्येक बिन्दु निम्न को संतुष्ट करता है

$$2\left(x^2+y^2\right)(x\,dx+y\,dy)=\left[\left(x^2+y^2+2\theta\right)^2-2\right](x\,dy-y\,dx).$$

वक्र C वृत्त $x^2+y^2=\dfrac{5}{3}$ से ठीक दो बिन्दुओं पर मिलता है। यदि इन बिन्दुओं पर θ के मान $\theta_1<\theta_2$ हैं, तो θ_1 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

With $r=\sqrt{x^2+y^2}$, $x=r\cos\theta$, $y=r\sin\theta$: $x\,dx+y\,dy=r\,dr$ and $x\,dy-y\,dx=r^2\,d\theta$.

Let $s=r^2$ (a second substitution). Then $x\,dx+y\,dy=r\,dr=\dfrac{1}{2}\,ds$, and along C, $s=g(\theta)$.

The equation becomes $2s\cdot\dfrac{1}{2}\,ds=\left[(s+2\theta)^2-2\right]s\,d\theta$. Since $s>0$, $\dfrac{ds}{d\theta}=(s+2\theta)^2-2$.

The right side is a function of the linear combination $s+2\theta$. Put $u=s+2\theta$: $\dfrac{du}{d\theta}=\dfrac{ds}{d\theta}+2=u^2$.

Separate: $\dfrac{du}{u^2}=d\theta$, so $-\dfrac{1}{u}=\theta+c$. At $\theta=0$, $s=1$, so $u=1$ and $c=-1$. Hence $u=\dfrac{1}{1-\theta}$ for $\theta<1$ and $s=g(\theta)=\dfrac{1}{1-\theta}-2\theta$.

Check that $s>0$ on $(-\pi,1)$: for $\theta\le 0$ both terms are positive; for $0<\theta<1$, $s'(\theta)=\dfrac{1}{(1-\theta)^2}-2$ vanishes at $1-\theta=\dfrac{1}{\sqrt{2}}$, where $s=2\sqrt{2}-2>0$. So C is well defined.

Intersection with $s=\dfrac{5}{3}$: $\dfrac{1}{1-\theta}=\dfrac{5}{3}+2\theta$. Multiplying by $3(1-\theta)$: $3=(5+6\theta)(1-\theta)=5+\theta-6\theta^2$, so $6\theta^2-\theta-2=0$.

Roots: $\theta=\dfrac{1\pm 7}{12}$, that is, $\theta=\dfrac{2}{3}$ and $\theta=-\dfrac{1}{2}$. Both lie in $(-\pi,1)$, so there are exactly two points.

Hence $\theta_1=-\dfrac{1}{2}=-0.5$.

Final Answer: -0.50

Question 16

English

Let $y=y(x)$, $x\ge 1$, be the solution of the differential equation $\dfrac{dy}{dx}=\dfrac{y}{x}+\dfrac{x^3}{y^2}$ with $y(1)=1$. Then the value of $y(22)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $y=y(x)$, $x\ge 1$, अवकल समीकरण $\dfrac{dy}{dx}=\dfrac{y}{x}+\dfrac{x^3}{y^2}$ का वह हल है जिसके लिए $y(1)=1$ है। तब $y(22)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The term $\dfrac{x^3}{y^2}$ has degree 1, so the equation is not homogeneous. The substitution $y=vx$ is still

useful because it removes the $\frac{y}{x}$ term.

With $y = vx$, $\frac{dy}{dx} = v + x\frac{dv}{dx}$ and $\frac{x^3}{y^2} = \frac{x^3}{x^2v^2} = \frac{x}{v^2}$. So $v + x\frac{dv}{dx} = v + \frac{x}{v^2}$.

This gives $\frac{dv}{dx} = \frac{1}{v^2}$, so $v^2\,dv = dx$ and $\frac{v^3}{3} = x + c_1$, that is, $v^3 = 3x + c$.

At $x = 1$: $v = \frac{y}{x} = 1$, so $1 = 3 + c$ and $c = -2$. Hence $v = (3x - 2)^{1/3}$, which is positive for $x \geq 1$, and $y = x(3x - 2)^{1/3}$.

At $x = 22$: $3 \cdot 22 - 2 = 64$, $v = 4$, $y = 88$.

Final Answer: 88

Question 17

English

Let $y = y(x)$, $x > 10$, be the solution of the differential equation

$$x + y\frac{dy}{dx} = \sqrt{x^2 + y^2}\left(1 + \frac{dy}{dx}\right)$$

with $y(15) = 20$. If $\alpha > 10$ is such that $y(\alpha) = \frac{21}{2}$, then the value of α is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $y = y(x)$, $x > 10$, अवकल समीकरण

$$x + y\frac{dy}{dx} = \sqrt{x^2 + y^2}\left(1 + \frac{dy}{dx}\right)$$

का वह हल है जिसके लिए $y(15) = 20$ है। यदि $\alpha > 10$ इस प्रकार है कि $y(\alpha) = \frac{21}{2}$, तो α का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write the equation in differentials: $x\,dx + y\,dy = \sqrt{x^2 + y^2}\,(dx + dy)$.

In polar form $x = r\cos\theta$, $y = r\sin\theta$, we have $x\,dx + y\,dy = \frac{1}{2}d\left(x^2 + y^2\right) = r\,dr$. So the equation becomes $r\,dr = r\,(dx + dy)$, and since $r > 0$, $dr = dx + dy$.

Integrate: $r = x + y + c$. At $(15, 20)$, $r = 25$, so $c = -10$: $\sqrt{x^2 + y^2} = x + y - 10$, with $x + y - 10 > 0$.

Square: $x^2 + y^2 = x^2 + y^2 + 100 + 2xy - 20x - 20y$, so $xy - 10x - 10y + 50 = 0$, that is, $(x - 10)(y - 10) = 50$. (The distance from the origin is $\sqrt{2}$ times the distance from the line $x + y = 10$, so the curve is a hyperbola.)

On $x > 10$ the solution is $y = 10 + \frac{50}{x - 10}$; at $x = 15$ it gives 20, as required.

$y(\alpha) = \frac{21}{2}$: $\frac{50}{\alpha - 10} = \frac{1}{2}$, so $\alpha = 110$. Check: $\sqrt{110^2 + 10.5^2} = 110.5 = 110 + 10.5 - 10$.

Final Answer: 110

Question 18

English

Let $y = y(x)$ be the solution of the differential equation

$$(x + 2y)\,(x\,dx + y\,dy) = (2x - y)\,(x\,dy - y\,dx)$$

with $y(1) = -3$, defined on an interval containing $[0, 1]$. Then the value of $y(0)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $y = y(x)$ अवकल समीकरण

$$(x + 2y)\,(x\,dx + y\,dy) = (2x - y)\,(x\,dy - y\,dx)$$

का वह हल है जिसके लिए $y(1) = -3$ है, तथा यह $[0, 1]$ को समाहित करने वाले अन्तराल पर परिभाषित है। तब $y(0)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Both sides are homogeneous of degree 2, and the combinations $x\,dx + y\,dy$ and $x\,dy - y\,dx$ suggest polar coordinates: $x\,dx + y\,dy = r\,dr$, $x\,dy - y\,dx = r^2\,d\theta$.

Also $x + 2y = r(\cos\theta + 2\sin\theta)$ and $2x - y = r(2\cos\theta - \sin\theta)$. The equation becomes $r^2(\cos\theta + 2\sin\theta)\,dr = r^3(2\cos\theta - \sin\theta)\,d\theta$.

Separate: $\dfrac{dr}{r} = \dfrac{2\cos\theta - \sin\theta}{\cos\theta + 2\sin\theta}\,d\theta$. The numerator is the derivative of the denominator, so $\ln r = \ln|\cos\theta + 2\sin\theta| + c$, and $r = k(\cos\theta + 2\sin\theta)$.

Multiply by r: $x^2 + y^2 = k(x + 2y)$, a family of circles through the origin.

At $(1, -3)$: $10 = k(1 - 6)$, so $k = -2$ and the circle is $x^2 + y^2 + 2x + 4y = 0$, with centre $(-1, -2)$ and radius $\sqrt{5}$.

As a function of x the solution through $(1, -3)$ is the lower arc $y = -2 - \sqrt{5 - (x + 1)^2}$ (it gives -3 at $x = 1$), defined for $-1 - \sqrt{5} < x < -1 + \sqrt{5}$, which contains $[0, 1]$. The upper arc passes through the origin, where the equation is singular.

Hence $y(0) = -2 - \sqrt{4} = -4$.

Final Answer: -4

Question 19

English

Let $y = y(x)$ be the solution of the differential equation $\dfrac{dy}{dx} = \dfrac{1 - y}{x + 2y - 3}$ with $y(3) = \dfrac{1}{2}$. The solution curve meets the line $x + y = 5$ at a point (a, b). Then the value of b is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $y = y(x)$ अवकल समीकरण $\dfrac{dy}{dx} = \dfrac{1 - y}{x + 2y - 3}$ का वह हल है जिसके लिए $y(3) = \dfrac{1}{2}$ है। हल वक्र रेखा $x + y = 5$ से बिन्दु (a, b) पर मिलता है। तब b का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The lines $1-y=0$ and $x+2y-3=0$ intersect at $(1,1)$. Shift the origin: $x=X+1$, $y=Y+1$. Then $1-y=-Y$ and $x+2y-3=X+2Y$, so $\dfrac{dY}{dX}=\dfrac{-Y}{X+2Y}$, which is homogeneous.

Put $Y=vX$: $v+X\dfrac{dv}{dX}=\dfrac{-v}{1+2v}$, so $X\dfrac{dv}{dX}=\dfrac{-v-v-2v^2}{1+2v}=-\dfrac{2v(1+v)}{1+2v}$.

Separate: $\dfrac{1+2v}{v(1+v)}\,dv=-\dfrac{2\,dX}{X}$. The numerator is the derivative of $v(1+v)$, so $\ln|v(1+v)|=-2\ln|X|+c$ and $v(1+v)X^2=K$.

Since $v(1+v)X^2=Y(X+Y)$, the solutions are $Y(X+Y)=K$, that is, $(y-1)(x+y-2)=K$.

At $(3,\frac{1}{2})$: $(-\frac{1}{2})(\frac{3}{2})=-\frac{3}{4}$, so $K=-\frac{3}{4}$. On this branch $Y<0$ throughout (it cannot reach $Y=0$ because $K\neq 0$), and $X=-\frac{3}{4Y}-Y$.

The line $x+y=5$ is $X+Y=3$. Then $3Y=-\frac{3}{4}$, $Y=-\frac{1}{4}$, and $X=3-Y=\frac{13}{4}$, so the point is $\left(\frac{17}{4},\frac{3}{4}\right)$; it is the only intersection.

Hence $b=0.75$.

Final Answer: 0.75

Question 20

English

Let $y=y(x)$, $0\le x\le 5$, be the solution of the differential equation $\dfrac{dy}{dx}=\dfrac{x+3y+19}{3x+y+9}$ with $y(0)=-6$. Then the value of $y(5)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $y=y(x)$, $0\le x\le 5$, अवकल समीकरण $\dfrac{dy}{dx}=\dfrac{x+3y+19}{3x+y+9}$ का वह हल है जिसके लिए $y(0)=-6$ है। तब $y(5)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Solve $x+3y+19=0$, $3x+y+9=0$: $(x,y)=(-1,-6)$. Put $x=X-1$, $y=Y-6$. Then $x+3y+19=X+3Y$ and $3x+y+9=3X+Y$, so $\dfrac{dY}{dX}=\dfrac{X+3Y}{3X+Y}$, which is homogeneous.

Put $Y=vX$: $v+X\dfrac{dv}{dX}=\dfrac{1+3v}{3+v}$, so $X\dfrac{dv}{dX}=\dfrac{1+3v-3v-v^2}{3+v}=\dfrac{1-v^2}{3+v}$.

Separate and use partial fractions: $\dfrac{3+v}{(1-v)(1+v)}=\dfrac{2}{1-v}+\dfrac{1}{1+v}$. Integrating, $-2\ln|1-v|+\ln|1+v|=\ln|X|+c$.

So $\dfrac{1+v}{(1-v)^2}=KX$. Multiply by X and use $v=\dfrac{Y}{X}$: $\dfrac{X(X+Y)}{(X-Y)^2}=KX$, that is, $X+Y=K(X-Y)^2$, a family of parabolas.

At $x=0$, $y=-6$: $X=1$, $Y=0$, so $K=1$ and $X+Y=(X-Y)^2$.

Parametrise by $s=X-Y$: $X+Y=s^2$, so $X=\dfrac{s^2+s}{2}$, $Y=\dfrac{s^2-s}{2}$. At the initial point $s=1$. Since $\dfrac{dX}{ds}=\dfrac{2s+1}{2}>0$ for $s>0$ and $3X+Y=2s^2+s>0$, the solution is a smooth function $y(x)$ for $s\ge 1$.

$x = 5$ means $X = 6$: $s^2 + s = 12$, $s = 3$, so $Y = \dfrac{9-3}{2} = 3$. (The other root $s = -4$ lies on a different branch.)
Hence $y(5) = Y - 6 = -3$.

Final Answer: -3

Question 21

English

Consider the differential equation $\dfrac{dy}{dx} = \dfrac{3x+y-1}{x+2y+3}$. For $\lambda \in \mathbb{R}$, let L_λ denote the line $(3x+y-1)+\lambda(x+2y+3)=0$. There are exactly two real values λ_1 and λ_2 of λ for which, at every point of L_λ with $x+2y+3 \neq 0$, the slope of L_λ is equal to the value of $\dfrac{dy}{dx}$ given by the differential equation. Then the value of $\lambda_1\lambda_2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

अवकल समीकरण $\dfrac{dy}{dx} = \dfrac{3x+y-1}{x+2y+3}$ पर विचार कीजिए। $\lambda \in \mathbb{R}$ के लिए माना कि L_λ रेखा $(3x+y-1)+\lambda(x+2y+3)=0$ को दर्शाता है। λ के ठीक दो वास्तविक मान λ_1 तथा λ_2 ऐसे हैं जिनके लिए L_λ के प्रत्येक बिन्दु पर, जहाँ $x+2y+3 \neq 0$ है, L_λ की प्रवणता अवकल समीकरण द्वारा दिए गए $\dfrac{dy}{dx}$ के मान के बराबर है। तब $\lambda_1\lambda_2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

All lines L_λ belong to the family through the point of intersection of $3x+y-1=0$ and $x+2y+3=0$, which is $(1,-2)$.
Shift the origin there: $x = X+1$, $y = Y-2$. Then $3x+y-1 = 3X+Y$ and $x+2y+3 = X+2Y$, so the equation becomes the homogeneous equation $\dfrac{dY}{dX} = \dfrac{3X+Y}{X+2Y}$.

The line L_λ is $(3+\lambda)X+(1+2\lambda)Y=0$, that is, $Y = mX$ with $m = -\dfrac{3+\lambda}{1+2\lambda}$ (for $\lambda = -\frac{1}{2}$ the line is $X=0$, which is not a solution since $X+2Y \neq 0$ there gives a finite slope).
Along $Y = mX$ the equation gives $\dfrac{dY}{dX} = \dfrac{3X+mX}{X+2mX} = \dfrac{3+m}{1+2m}$, a constant. So L_λ is a solution curve exactly when $m = \dfrac{3+m}{1+2m}$, that is, $m+2m^2 = 3+m$, $m^2 = \dfrac{3}{2}$, $m = \pm\sqrt{\dfrac{3}{2}}$.

From $m(1+2\lambda) = -(3+\lambda)$: $\lambda(2m+1) = -(3+m)$, so $\lambda = -\dfrac{3+m}{2m+1}$, which gives one λ for each m; these are $\lambda = \pm\dfrac{\sqrt{6}}{2}$.

Product: $\lambda_1\lambda_2 = \dfrac{(3+m_1)(3+m_2)}{(1+2m_1)(1+2m_2)}$ with $m_1+m_2=0$ and $m_1m_2 = -\dfrac{3}{2}$: numerator $9+3(m_1+m_2)+m_1m_2 = \dfrac{15}{2}$, denominator $1+2(m_1+m_2)+4m_1m_2 = -5$.

Hence $\lambda_1\lambda_2 = -\dfrac{3}{2} = -1.5$.

Final Answer: -1.50

Question 22

English

Let $f : (0, \infty) \to \mathbb{R}$ be a differentiable function with $f(1) = 2$ such that $xf'(x) = 2f(x) - x$ for all $x > 0$. Let C be the curve in the region $y > 0$ passing through the point $\left(\frac{3}{2}, 1\right)$ such that every point (x, y) of C satisfies $y\,dx = (x + y\,f(y))\,dy$. If the point $(\alpha, 6)$ lies on C, then the value of α is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : (0, \infty) \to \mathbb{R}$ एक अवकलनीय फलन है, $f(1) = 2$, जिसके लिए सभी $x > 0$ हेतु $xf'(x) = 2f(x) - x$ है। माना कि C क्षेत्र $y > 0$ में बिन्दु $\left(\frac{3}{2}, 1\right)$ से होकर जाने वाला वह वक्र है जिसका प्रत्येक बिन्दु (x, y) समीकरण $y\,dx = (x + y\,f(y))\,dy$ को संतुष्ट करता है। यदि बिन्दु $(\alpha, 6)$, C पर स्थित है, तो α का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

First equation: $f'(x) - \frac{2}{x}f(x) = -1$, linear in f with $P = -\frac{2}{x}$; integrating factor $e^{-2\ln x} = \frac{1}{x^2}$.

So $\frac{d}{dx}\left(\frac{f}{x^2}\right) = -\frac{1}{x^2}$ and $\frac{f(x)}{x^2} = \frac{1}{x} + c$. With $f(1) = 2$: $2 = 1 + c$, $c = 1$. Hence $f(x) = x + x^2$.

Curve: $y\,dx = \big(x + y(y^2 + y)\big)\,dy$. As $\frac{dy}{dx} = \frac{y}{x + y^3 + y^2}$ it is not linear in y, but with x as a function of y: $\frac{dx}{dy} = \frac{x}{y} + y^2 + y$, which is linear in x.

Integrating factor $e^{-\int dy/y} = \frac{1}{y}$: $\frac{d}{dy}\left(\frac{x}{y}\right) = y + 1$, so $\frac{x}{y} = \frac{y^2}{2} + y + K$.

At $\left(\frac{3}{2}, 1\right)$: $\frac{3}{2} = \frac{1}{2} + 1 + K$, so $K = 0$ and $x = \frac{y^3}{2} + y^2$.

At $y = 6$: $\alpha = \frac{216}{2} + 36 = 144$.

Final Answer: 144

Question 23

English

Let $y : (0, \infty) \to \mathbb{R}$ be a differentiable function such that $x\dfrac{dy}{dx} = y - x^3y^2$ for all $x > 0$ and $y(1) = \dfrac{4}{5}$. Then the value of $y(2)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $y : (0, \infty) \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए सभी $x > 0$ हेतु $x\dfrac{dy}{dx} = y - x^3y^2$ तथा $y(1) = \dfrac{4}{5}$ है। तब $y(2)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Divide by x: $\dfrac{dy}{dx} - \dfrac{1}{x}y = -x^2y^2$, a Bernoulli equation with $n = 2$.

Since $y(1) \neq 0$ and the solution we obtain never vanishes, put $w = \dfrac{1}{y}$: $\dfrac{dw}{dx} = -\dfrac{1}{y^2}\dfrac{dy}{dx}$. Dividing the equation by $-y^2$: $\dfrac{dw}{dx} + \dfrac{1}{x}w = x^2$.

This is linear in w with integrating factor x: $\dfrac{d}{dx}(xw) = x^3$, so $xw = \dfrac{x^4}{4} + c$.

At $x = 1$: $w = \dfrac{5}{4}$, so $\dfrac{5}{4} = \dfrac{1}{4} + c$, $c = 1$. Hence $w = \dfrac{x^4 + 4}{4x}$ and $y = \dfrac{4x}{x^4 + 4}$, which is positive on $(0, \infty)$.

$y(2) = \dfrac{8}{16 + 4} = 0.4$.

Final Answer: 0.40

Question 24

English

Let $y : (0, \infty) \to (0, \infty)$ be a differentiable function such that $2xy\dfrac{dy}{dx} - y^2 = x^2$ for all $x > 0$ and $y(1) = 2$. Then the value of $y\left(\dfrac{12}{5}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $y : (0, \infty) \to (0, \infty)$ एक अवकलनीय फलन है जिसके लिए सभी $x > 0$ हेतु $2xy\dfrac{dy}{dx} - y^2 = x^2$ तथा $y(1) = 2$ है। तब $y\left(\dfrac{12}{5}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Divide by $2xy$: $\dfrac{dy}{dx} - \dfrac{1}{2x}y = \dfrac{x}{2}y^{-1}$, a Bernoulli equation with $n = -1$.

The substitution $u = y^{1-n} = y^2$ makes it linear: $\dfrac{du}{dx} = 2y\dfrac{dy}{dx}$, and the original equation multiplied out reads $x\dfrac{du}{dx} - u = x^2$, that is, $\dfrac{du}{dx} - \dfrac{u}{x} = x$.

Integrating factor $\dfrac{1}{x}$: $\dfrac{d}{dx}\left(\dfrac{u}{x}\right) = 1$, so $\dfrac{u}{x} = x + c$ and $u = x^2 + cx$.

At $x = 1$: $u = 4$, so $c = 3$. Since $y > 0$, $y = \sqrt{x^2 + 3x}$.

At $x = \dfrac{12}{5}$: $x^2 + 3x = \dfrac{144}{25} + \dfrac{180}{25} = \dfrac{324}{25}$, so $y = \dfrac{18}{5} = 3.6$.

Final Answer: 3.60

Question 25

English

Let $y : (0, \infty) \to (0, \infty)$ be a differentiable function such that $x\dfrac{dy}{dx} = y\,(x + \ln y)$ for all $x > 0$ and $y(1) = 1$. Then the value of $y(3)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $y : (0, \infty) \to (0, \infty)$ एक अवकलनीय फलन है जिसके लिए सभी $x > 0$ हेतु $x\dfrac{dy}{dx} = y\,(x + \ln y)$ तथा $y(1) = 1$ है। तब $y(3)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Divide by xy (both positive): $\dfrac{1}{y}\dfrac{dy}{dx} = 1 + \dfrac{\ln y}{x}$.

The left side is $\dfrac{d}{dx}(\ln y)$, so put $u = \ln y$: $\dfrac{du}{dx} - \dfrac{1}{x}u = 1$, a linear equation in u.

Integrating factor $\dfrac{1}{x}$: $\dfrac{d}{dx}\left(\dfrac{u}{x}\right) = \dfrac{1}{x}$, so $\dfrac{u}{x} = \ln x + c$.

At $x = 1$: $u = \ln 1 = 0$, so $c = 0$ and $u = x \ln x$. Hence $y = e^{x \ln x} = x^x$.

$y(3) = 3^3 = 27$.

Final Answer: 27

Question 26

English

Let $y = y(x)$, $2 \le x \le \dfrac{11}{4}$, be the solution of the differential equation

$$\left(y^3 + 2xy\right) dx + \left(xy^2 - x^2\right) dy = 0$$

with $y(2) = 8$. Then the value of $y\left(\dfrac{11}{4}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $y = y(x)$, $2 \le x \le \dfrac{11}{4}$, अवकल समीकरण

$$\left(y^3 + 2xy\right) dx + \left(xy^2 - x^2\right) dy = 0$$

का वह हल है जिसके लिए $y(2) = 8$ है। तब $y\left(\dfrac{11}{4}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

With $M = y^3 + 2xy$ and $N = xy^2 - x^2$: $\dfrac{\partial M}{\partial y} = 3y^2 + 2x$ and $\dfrac{\partial N}{\partial x} = y^2 - 2x$, so the equation is not exact.

Dividing by y^2 (an integrating factor found by inspection): $\left(y + \dfrac{2x}{y}\right) dx + \left(x - \dfrac{x^2}{y^2}\right) dy = 0$. Group

the terms: $(y\,dx + x\,dy) + \left(\frac{2x}{y}dx - \frac{x^2}{y^2}dy\right) = d(xy) + d\left(\frac{x^2}{y}\right)$.

So the equation is $d\left(xy + \frac{x^2}{y}\right) = 0$ and the solution is $xy + \frac{x^2}{y} = c$. At $(2, 8)$: $c = 16 + \frac{4}{8} = \frac{33}{2}$.

Multiply by y: $xy^2 - \frac{33}{2}y + x^2 = 0$, so $y = \frac{\frac{33}{2} \pm \sqrt{\frac{1089}{4} - 4x^3}}{2x}$.

At $x = 2$: the square root is $\sqrt{\frac{1089}{4} - 32} = \frac{31}{2}$, giving $y = 8$ or $\frac{1}{4}$; the solution uses the $+$ sign. On $\left[2, \frac{11}{4}\right]$, $\frac{1089}{4} - 4x^3 \geq \frac{1089}{4} - \frac{1331}{16} > 0$, so the $+$ branch is continuous there and $xy^2 - x^2 \neq 0$ along it.

At $x = \frac{11}{4}$: $\sqrt{\frac{1089}{4} - \frac{1331}{16}} = \sqrt{\frac{3025}{16}} = \frac{55}{4}$, so $y = \frac{\frac{33}{2} + \frac{55}{4}}{\frac{11}{2}} = \frac{121}{4} \cdot \frac{2}{11} = \frac{11}{2} = 5.5$.

Final Answer: 5.50

Question 27

English

Consider the family of curves $y = \frac{c(x+1)}{x}$, $x \neq 0$, where c is a real parameter. The orthogonal trajectory of this family that passes through the point $(12, 36)$ is part of a curve Γ with a polynomial equation. The curve Γ meets the x-axis at the origin and at exactly one more point $(\alpha, 0)$. Then the value of α is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वक्र-कुल $y = \frac{c(x+1)}{x}$, $x \neq 0$ पर विचार कीजिए, जहाँ c एक वास्तविक प्राचल है। इस कुल का बिन्दु $(12, 36)$ से होकर जाने वाला लंबकोणीय संछेदी (orthogonal trajectory) एक वक्र Γ का भाग है जिसका समीकरण बहुपदीय है। वक्र Γ, x-अक्ष से मूलबिन्दु पर तथा ठीक एक अन्य बिन्दु $(\alpha, 0)$ पर मिलता है। तब α का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Differentiate $y = c\left(1 + \frac{1}{x}\right)$: $y' = -\frac{c}{x^2}$. Eliminate $c = \frac{xy}{x+1}$: $y' = -\frac{y}{x(x+1)}$. This is the differential equation of the family.

For the orthogonal trajectories replace y' by $-\frac{1}{y'}$: $y' = \frac{x(x+1)}{y}$.

Separate: $y\,dy = (x^2 + x)\,dx$, so $\frac{y^2}{2} = \frac{x^3}{3} + \frac{x^2}{2} + K$.

At $(12, 36)$: $\frac{1296}{2} = 648$ and $\frac{1728}{3} + \frac{144}{2} = 576 + 72 = 648$, so $K = 0$. The trajectory lies on $\Gamma : 3y^2 = 2x^3 + 3x^2 = x^2(2x + 3)$.

On the x-axis: $x^2(2x + 3) = 0$, so $x = 0$ or $x = -\frac{3}{2}$. Hence $\alpha = -1.5$.

Check of orthogonality at $(12, 36)$: family slope $-\frac{36}{12 \cdot 13} = -\frac{3}{13}$, trajectory slope $\frac{12 \cdot 13}{36} = \frac{13}{3}$; product -1.

Final Answer: -1.50

Question 28

English

Consider the family of all circles that pass through the origin and have their centres on the line $y = x$. Let S be the orthogonal trajectory of this family that passes through the point $(1, 3)$. The curve S is part of a circle which meets the x-axis at the origin and at one more point $(\beta, 0)$. Then the value of β is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

उन सभी वृत्तों के कुल पर विचार कीजिए जो मूलबिन्दु से होकर जाते हैं तथा जिनके केन्द्र रेखा $y = x$ पर हैं। माना कि S इस कुल का बिन्दु $(1, 3)$ से होकर जाने वाला लंबकोणीय संछेदी है। वक्र S एक वृत्त का भाग है जो x-अक्ष से मूलबिन्दु पर तथा एक अन्य बिन्दु $(\beta, 0)$ पर मिलता है। तब β का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A circle through the origin with centre (c, c) is $x^2+y^2 = 2c(x+y)$. Differentiate: $2x+2yp = 2c(1+p)$, where $p = \dfrac{dy}{dx}$, so $c = \dfrac{x + yp}{1 + p}$.

Eliminate c: $(x^2 + y^2)(1 + p) = 2(x + y)(x + yp)$. Solving for p: $p\left(x^2 + y^2 - 2xy - 2y^2\right) = 2x^2 + 2xy - x^2 - y^2$, so $p = \dfrac{x^2 + 2xy - y^2}{x^2 - 2xy - y^2}$ is the slope of the family; this is its differential equation.

For the orthogonal trajectories replace p by $-\dfrac{1}{p}$: $\dfrac{dy}{dx} = -\dfrac{x^2 - 2xy - y^2}{x^2 + 2xy - y^2}$.

This equation is homogeneous; by the same elimination it is the differential equation of the circles $x^2 + y^2 = 2k(x - y)$ (centres $(k, -k)$ on $y = -x$): differentiating gives $k = \dfrac{x + yp}{1 - p}$ and eliminating k gives $(x^2 + y^2)(1 - p) = 2(x - y)(x + yp)$, which rearranges to the equation above.

Through $(1, 3)$: $1 + 9 = 2k(1 - 3)$, so $k = -\dfrac{5}{2}$ and the circle is $x^2 + y^2 + 5x - 5y = 0$.

On the x-axis: $x^2 + 5x = 0$, so $x = 0$ or $x = -5$. Hence $\beta = -5$.

Check of orthogonality at $(1, 3)$: the family member through $(1, 3)$ has $c = \dfrac{10}{8}$, centre $\left(\frac{5}{4}, \frac{5}{4}\right)$; the trajectory has centre $\left(-\frac{5}{2}, \frac{5}{2}\right)$. The radii to $(1, 3)$ are $\left(-\frac{1}{4}, \frac{7}{4}\right)$ and $\left(\frac{7}{2}, \frac{1}{2}\right)$, with dot product $-\frac{7}{8} + \frac{7}{8} = 0$, so the circles cut at right angles.

Final Answer: -5

Question 29

English

Let $y = f(x)$, $0 \le x \le 7$, be a differentiable function with $f(0) = 2$, $f(x) > 0$ and $f'(x) > 0$ for all x. For each point P on the curve $y = f(x)$, let T be the point where the tangent at P meets the x-axis and let M be the foot of the perpendicular from P to the x-axis. If the area of triangle PMT is 8 for every such P, then the value of $f(7)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $y = f(x)$, $0 \le x \le 7$, एक अवकलनीय फलन है जिसके लिए $f(0) = 2$ तथा सभी x हेतु $f(x) > 0$ एवं $f'(x) > 0$ है। वक्र $y = f(x)$ के प्रत्येक बिन्दु P के लिए माना कि T वह बिन्दु है जहाँ P पर स्पर्श रेखा x-अक्ष से मिलती है तथा M, P से x-अक्ष पर डाले गए लम्ब का पाद है। यदि ऐसे प्रत्येक P के लिए त्रिभुज PMT का क्षेत्रफल 8 है, तो $f(7)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

At $P(x, y)$ the tangent is $Y - y = y'(X - x)$; it meets the x-axis at $T\left(x - \dfrac{y}{y'}, 0\right)$, and $M = (x, 0)$.

So $PM = y$ and $MT = \dfrac{y}{y'}$ (the subtangent; positive since $y, y' > 0$).

Triangle PMT is right-angled at M, so its area is $\dfrac{1}{2} \cdot y \cdot \dfrac{y}{y'} = \dfrac{y^2}{2y'}$. The condition gives $\dfrac{y^2}{2y'} = 8$, that is, $\dfrac{dy}{dx} = \dfrac{y^2}{16}$.

Separate: $\dfrac{dy}{y^2} = \dfrac{dx}{16}$, so $-\dfrac{1}{y} = \dfrac{x}{16} + c$.

$f(0) = 2$: $-\dfrac{1}{2} = c$. Hence $\dfrac{1}{y} = \dfrac{1}{2} - \dfrac{x}{16} = \dfrac{8 - x}{16}$ and $y = \dfrac{16}{8 - x}$, which is positive and increasing on $[0, 7]$.

$f(7) = \dfrac{16}{1} = 16$.

Final Answer: 16

Question 30

English

A tank initially contains 100 litres of brine in which 2 kg of salt is dissolved. Pure water flows into the tank at the rate of 1 litre per minute, and the well-stirred mixture flows out at the rate of 2 litres per minute. Then the amount of salt, in kg, in the tank at the instant when the tank contains 60 litres of brine is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक टंकी में प्रारम्भ में 100 लीटर लवण-जल है जिसमें 2 kg लवण घुला है। टंकी में शुद्ध जल 1 लीटर प्रति मिनट की दर से प्रवेश करता है तथा भली-भाँति मिश्रित घोल 2 लीटर प्रति मिनट की दर से बाहर निकलता है। तब जिस क्षण टंकी में 60 लीटर लवण-जल है, उस क्षण टंकी में लवण की मात्रा, kg में, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $S(t)$ kg be the salt and $V(t)$ litres the volume at time t minutes. The net outflow is 1 litre per minute, so $V = 100 - t$.

Pure water brings no salt; the outflow removes salt at the rate $2 \cdot \dfrac{S}{V}$ kg per minute. So $\dfrac{dS}{dt} = -\dfrac{2S}{100 - t}$, $0 \le t < 100$.

Separate: $\dfrac{dS}{S} = -\dfrac{2\,dt}{100 - t}$, so $\ln S = 2\ln(100 - t) + c$ and $S = k(100 - t)^2$.

$S(0) = 2$: $k = \dfrac{2}{10^4}$, so $S = 2\left(\dfrac{100 - t}{100}\right)^2 = 2\left(\dfrac{V}{100}\right)^2$.

When $V = 60$ ($t = 40$): $S = 2 \cdot (0.6)^2 = 0.72$.

A constant-volume model would give the exponential $2e^{-2t/100}$ instead, a common error.

Final Answer: 0.72

Question 31

English

Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function such that $3\int_0^x f(t)\,dt = x\,(f(x) + 2f(0))$ for all $x \in \mathbb{R}$, $f(0) = 2$ and $f(1) = 5$. Then the value of $f(8)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए सभी $x \in \mathbb{R}$ हेतु $3\int_0^x f(t)\,dt = x\,(f(x) + 2f(0))$, $f(0) = 2$ तथा $f(1) = 5$ है। तब $f(8)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The left side is differentiable with derivative $3f(x)$ (Newton-Leibniz). Differentiate both sides: $3f(x) = f(x) + 2f(0) + xf'(x)$, so $xf'(x) = 2\,(f(x) - f(0))$.

Put $g(x) = f(x) - f(0)$: $xg'(x) = 2g(x)$. On $(0, \infty)$ this separates: $\dfrac{g'}{g} = \dfrac{2}{x}$ where $g \neq 0$, and every solution is $g = kx^2$. So $f(x) = 2 + kx^2$ for $x > 0$.

Check in the integral equation for $x > 0$: $3\displaystyle\int_0^x \left(2 + kt^2\right) dt = 6x + kx^3$ and $x\left(2 + kx^2 + 4\right) = 6x + kx^3$, equal for every k. So the equation itself does not fix k; the value $f(1) = 5$ does: $2 + k = 5$, $k = 3$.

Hence $f(x) = 2 + 3x^2$ for $x > 0$ and $f(8) = 2 + 3 \cdot 64 = 194$. (The values for $x < 0$ do not affect $f(8)$.)

Final Answer: 194

Question 32

English

Let $y : \mathbb{R} \to \mathbb{R}$ be a differentiable function such that $\frac{dy}{dx} = y - x^2$ for all $x \in \mathbb{R}$ and $\lim_{x\to\infty} \frac{y(x)}{x^3}$ exists as a real number. Then the value of $y(6)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $y : \mathbb{R} \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए सभी $x \in \mathbb{R}$ हेतु $\frac{dy}{dx} = y - x^2$ है तथा $\lim_{x\to\infty} \frac{y(x)}{x^3}$ एक वास्तविक संख्या के रूप में विद्यमान है। तब $y(6)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write the equation as $y' - y = -x^2$, linear with $P = -1$; integrating factor e^{-x}.

$\frac{d}{dx}(ye^{-x}) = -x^2e^{-x}$. Integrating by parts twice, $\int -x^2e^{-x}dx = (x^2 + 2x + 2)e^{-x} + C$.

So $y = x^2 + 2x + 2 + Ce^x$. Every real C gives a solution on $\mathbb{R}$; a condition is needed to fix C. Here it is a boundary condition at infinity.

$\frac{y}{x^3} = \frac{x^2 + 2x + 2}{x^3} + C\frac{e^x}{x^3}$. The first part tends to 0; $\frac{e^x}{x^3} \to \infty$. So the limit is a real number only when $C = 0$ (it is $+\infty$ for $C > 0$ and $-\infty$ for $C < 0$).

Hence $y = x^2 + 2x + 2$ and $y(6) = 36 + 12 + 2 = 50$.

Final Answer: 50